https://k8s-school.fr

Docker par la pratique

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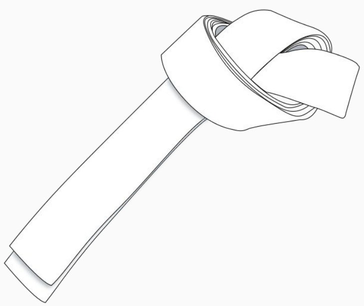
Exercice 1.1

Start a container named "mariadbtest" and run a mariadb instance



MariaDB

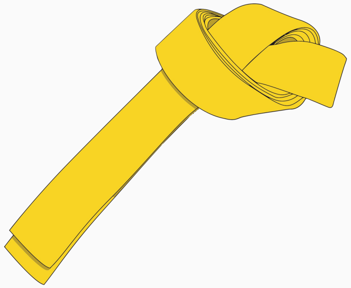
Solution

[www.docker.com]

```
# mariadbtest is the name we want to assign the container. If we don't specify a name, an id will be automatically generated.
docker run --name mariadbtest -e MYSQL_ROOT_PASSWORD=mypass -d mariadb
# Optionally, after the image name, we can specify some options for mysqld. For example:
docker run --name mariadbtest -e MYSQL_ROOT_PASSWORD=mypass -d mariadb --log-bin
--binlog-format=MIXED
# list running containers
docker ps
```

Exercice 1.2

Install vim into mariadbtest container



Solution

[www.docker.com]

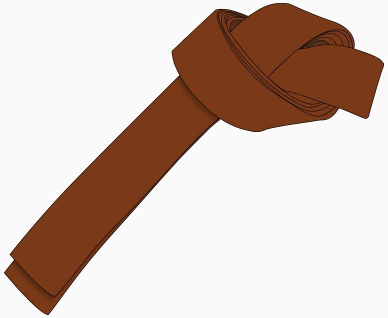
```
# Access the container via bash, with root access
docker exec -it mariadbtest bash

# Install stuff inside the container
apt-get install vim
```

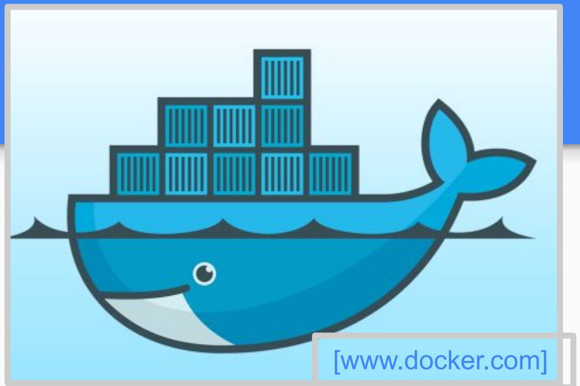
Exercice 2

Create a Dockerfile to run a python webserver, use:

cd k8s-school/docker/webserver



Solution




```
# Use ubuntu as base image, it will be downloaded automatically
FROM ubuntu:latest
MAINTAINER Fabrice Jammes "fabrice.jammes@gmail.com"

# Update and install system dependencies
RUN apt-get update && apt-get install -y python3

RUN mkdir -p /home/www
WORKDIR /home/www

# Commands below will all be launched in WORKDIR

# Launch command below at container startup
# It will serve files located where it has been launch
# so /home/www
CMD python3 /home/src/hello.py

# Add local file inside container
# code can also be retrieved from git repository
ADD index.html /home/www/index.html

# This command is the last one so, if hello.py is changed
# only this container layer will be changed.
ADD hello.py /home/src/hello.py
```

Solution

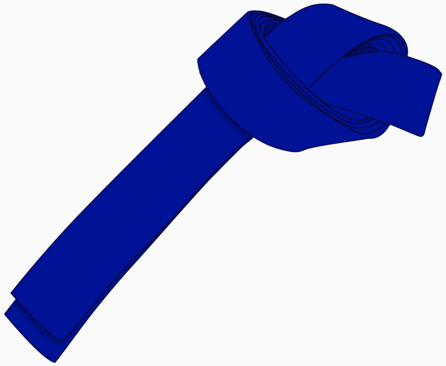
[www.docker.com]

\# Build image:

```
docker build --tag=webserver webserver/
```

Exercice 2

Launch container as a daemon and publish port 8000 on host



Solution

[www.docker.com]

```
# Option:
# --detach=true: run the container in the background and print the new container ID
# --publish: publish container's 8000 port to the host.
```

Exercice 2

Launch container as a daemon and publish port 8000 on host, and use html file stored on host machine

Solution

[www.docker.com]

```
docker run --name my_webserver -p 8000:8000 -v $PWD/www:/home/www webserver
```